

SCIENCE & MEASUREMENT

- **SCIENCE**

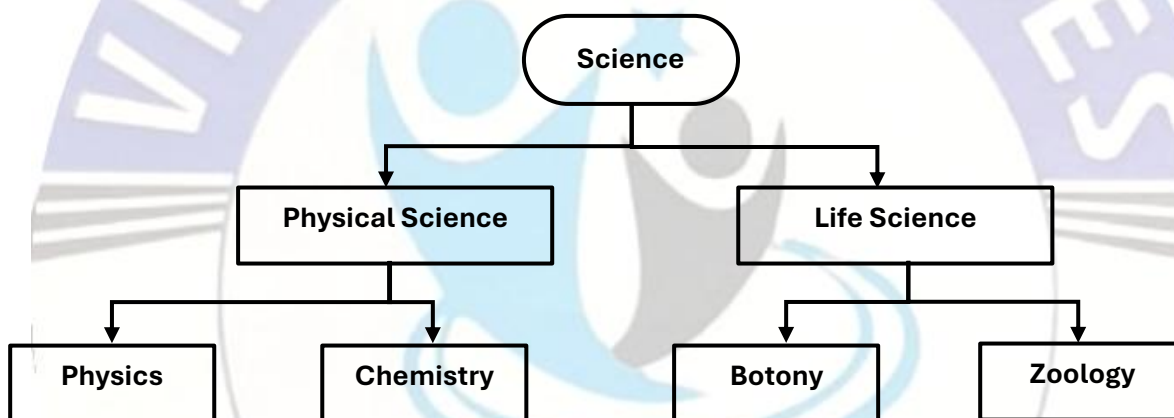
- Science is the pursuit and application of knowledge and understanding of the natural and social world following a systematic methodology based on evidence.

OR

The study of knowledge about the physical world and natural laws is called science.

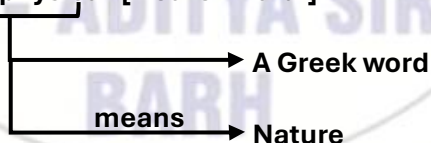
- The full form of science

S → Systematic
C → Comprehensive
I → Investigation
E → Exploration of
N → Natural
C → Causes &
E → Effect



- **Physics**

- The world of physics is derived from “physica” [Neuter Plural].



- Physics was also known as “Natural Philosophy” until the last 18th century.

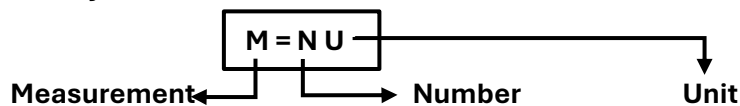
Note:

- “Aristotle” gave the term “physics” around 350 B.C.

- Physics helps us to understand how the world around us.
- Physics is essential to understanding our modern technological society.
- Physics develops our skills to understand the universe.
- Physics is also called “fundamental science”.
- The father of physics – “Sir Issac Newton”.

- Definition of physics

- **Physics is a science of measurement to measure means attached to a number and unit. It is mathematically written as**



Measurement can be expressed by several manners with the help of different numbers and units, but measurement can't be change.

i.e, $N_1 U_1 = N_2 U_2$

In this case numbers and units are inversely proportional to $N \propto \frac{1}{U}$.

For example:-

i) $1 \text{ meter} = 1 \times \text{meter}$ $\Rightarrow 1 \text{ meter} = 100 \times \text{cm}$

ii) $1 \text{ foot} = 1 \times \text{feet}$ $\Rightarrow 1 \text{ foot} = 12 \times \text{inch}$

- General definition of physics

- **Physics is a fundamental branch of science which deals with properties and behaviour of matter and energy.**

OR

Physics is the scientific study of physical phenomena like the motion of matter and energy and forces.

- Unit

- The quantity is to be measured by that standard scale, this standard scale is known as unit of that quantity.

OR

It is a standard quantity of the same kind with which a physical quantity is compared for measuring it.

For example:-

- i) The S.I unit of length is 'meter'.
- ii) The S.I unit of Mass is 'Kg'.
- iii) The S.I unit of time is 'second'.

Note:

- **S.I → International System of Unit.**

OR

Standard System of international Unit.

- **Magnitude**

- The numerical value of a measurement is called magnitude.
- It can be positive, negative or zero.

For example:-

- i) $15 \times 0 = 0 \times 15 = 0$ [Zero]
- ii) $15 \times 1 = 1 \times 15 = 15$ [+ve]
- iii) $-15 \times 2 = 2 \times (-15) = -30$ [-ve]

- **Physical quantity**

- The laws of physics are expressed in those terms are called physical quantities.

For example:-

- ✓ Length, Mass, Time, Velocity, Force, Work etc.

Types of physical quantity

- There are two types of physical quantity.

1. Base/Basic/Fundamental quantity.
2. Derived quantity.

1. **Base/Basic/Fundamental quantity**

- Those physical quantities which do not depend on other physical quantities are called basic quantities.
- There are seven basic quantities which are: Length, Mass, Time, Electric current, Luminous intensity, Amount of substance & temperature.

2. **Derived quantity**

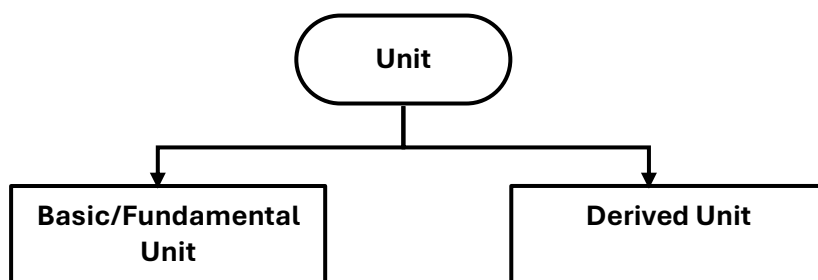
- Those physical quantities which depend on basic/fundamental quantities are called derived quantities.

For example:-

- ✓ Area, Volume, Force, Work, Energy etc.

- Now we expressed derived quantities by helping basic quantity.

Physical Quantity	Basic quantity used in derivation	Dimensional Formula
Area	$l \times b = L \times L$	$[L^2]$
Volume	$l \times b \times h = L \times L \times L$	$[L^3]$
Force	$m \times a = M \times LT^{-2}$	$[MLT^{-2}]$
Velocity	$\frac{S}{t} = \frac{L}{T}$	$[LT^{-1}]$
Acceleration	$\frac{V}{t} = \frac{LT^{-1}}{T}$	$[LT^{-2}]$
Work	Force $\times$ displacement = $F \times S$	$[ML^2T^{-2}]$
Energy	Equal to work	$[ML^2T^{-2}]$
Power	$\frac{W}{t}$	$[ML^2T^{-3}]$



1. Basic/Fundamental Unit

- Those units which do not depend on other units are called basic units.
- The units of basic quantities are called basic units.

i.e, there are seven basic units.

Basic quantity	Basic units	Symbol
Length	Meter	m
Time	Second	S
Mass	Kilogram	Kg
Electric Current	Ampere	A
Temperature	Kelvin	K
Luminous intensity	Candela	Cd
Amount of Substance	Mole	mol

2. Derived Units

- Those units which depends on basic units are called derived unit.
- The units of derived quantity are called derived units.

For example:-

- i) The S.I unit of velocity = m/s.
- ii) The S.I unit of acceleration = m/s².
- iii) The S.I unit of force = kg m/s².

• The system of some basic units

1. Foot – pound second system

- In this system, the unit of length is “foot”, the unit of mass is “pound” and the unit of time is “second”.
- This system is also known as British system.
- This system is also called F.P.S system in ‘Short’.

2. Centimeter – gram – second system

- In this system, the units of length is “cm”, the unit of mass is “gram” and the unit of time is “second”.
- It is called Metric System.
- It is also called C.G.S system in short.

3. Meter – Kilogram Second System

- In this system, the unit of length is “meter”, the unit of mass is “kg” and the unit of time is “second”.

- It is also called M.K.S system in short.

There are two other physical quantities

1. Vector quantity.
2. Scalar quantity.

1. Vector quantity

- Those physical quantities which have magnitudes as well as direction are called vector quantities.
- It is represented by the bold capital letter Or, arrow ($\rightarrow$) on the letters.

For example:-

- ✓ Force ($\vec{F}$), Velocity ($\vec{V}$), acceleration ($\vec{a}$), displacement ($\vec{s}$) etc.

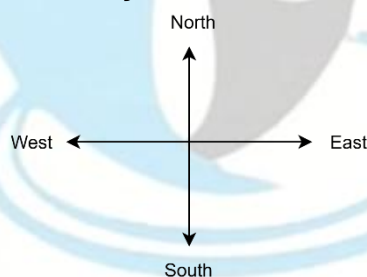
Note:

1. The length of the arrow gives magnitude.

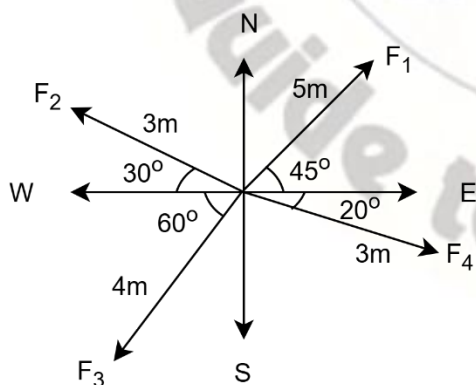


2. The head of arrow gives the direction of vector.

Only for understand



Q.



Ans Here,

$\vec{F}_1 = 5\text{m}$ and 45° from east (east to north).

$\vec{F}_2 = 3\text{m}$ and 30° from west (west to north).

$\vec{F}_3 = 4\text{m}$ and 60° from west (west to south).

$\vec{F}_4 = 3\text{m}$ and 20° from east (east to south).

- **Scalar quantities**

- Those physical quantities which have only magnitude are called Scalar quantities.

For example:-

- ✓ Area, Volume, density, energy, work, time, mass etc.

Difference between Vector & Scalar

Scalar	Vector
A Scalar is a quantity that has magnitude only, but no direction.	A vector is a quantity that has magnitude as well as direction both.
It follows normal rule (+, -, x, %) of algebra.	It follows vector laws.
Scalar quantity can change in terms of its magnitude only.	It can change in term of either in magnitude or direction.
It resultant is always Scalar.	It resultant is either vector or Scalar.

- **Prefix**

- These are the letters attached before a unit to make a larger or, smaller unit.

- These are summarized as follows:-

- **Case 1: Smaller prefix**

Prefix	Symbol	Magnitude & Value
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}
Femto	f	10^{-15}
Atto	a	10^{-18}

- **Case 2: Greater prefix**

Prefix	Symbol	Magnitude & Value
Kilo	K	10^3
Mega	M	10^6
Giga	G	10^9
Tera	T	10^{12}
Peta	P	10^{15}
Exa	E	10^{18}

Trick:

1. मै Micro 10^{-6}	ने Nano 10^{-9}	पीटा Peco 10^{-12}	फिर Femto 10^{-15}	आपको Atto 10^{-18}
2. Elephant Exa 10^{18}	पिये Peta 10^{15}	तो Tera 10^{15}	गधा Giga 10^9	मरे Mega 10^6

Note:

1 Millimeter (1mm) = 10^{-3} m
 1 Kilometer (1 Km) = 10^3 m

• **Scientific Notation**

▪ **Case 1:**

- If decimal is shifted towards left side then power of 10 increases.

For example:-

Number	Scientific Notation
235 = 235.0	2.35×10^2
476 = 476.0	4.76×10^2
16479	1.6479×10^4
24694	2.4694×10^4
6458134	6.458134×10^6

▪ **Case 2:**

- If decimal is shifted towards right side then power of 10 decreases.

Number	Scientific Notation
64.964	6496.4×10^{-2}
246.243	24624.3×10^{-2}
4941.24	49412.4×10^{-1}
249.645	24964.5×10^{-2}
$0.4 = \frac{4}{10}$	4×10^{-1}
$0.0003 = \frac{3}{10000}$	3×10^{-4}
0.0000241	2.41×10^{-5}

Note:

- Scientific Notation make the calculations involving very small and very large numbers very easy.

- **Temperature**

- It is a measurement of the hotness and coldness of a body.
- Temperature is measured in:-
 1. Degree Celcius.
 2. Fahrenheit.
 3. Kelvin.

- **Relationship between Celcius and kelvin**

$$\Rightarrow 0^{\circ}\text{C} = 273.15 \text{ Kelvin}$$
$$\Rightarrow 0^{\circ}\text{C} \approx 273\text{K (Approx)}$$

- **Relationship between Celcius and Fahrenheit**

$$\Rightarrow \text{Fahrenheit} = \frac{9}{5} \times ^{\circ}\text{C} + 32$$

- **Relationship between Celcius, Kelvin and Fahrenheit**

$$\Rightarrow \frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273}{5}$$

For example:-

Q. Calculate 45°C in (i) Kelvin, (ii) Fahrenheit.

Ans

(i) $\Rightarrow \frac{C}{5} = \frac{K - 273}{5}$

$$\Rightarrow C = K - 273$$

$$\Rightarrow 45^{\circ}\text{C} = K - 273$$

$$\Rightarrow 45 + 273 = 318 \text{ K}$$

$$\Rightarrow 45^{\circ}\text{C} = 318 \text{ K}$$

(ii) $\Rightarrow F = \frac{9}{5} \times ^{\circ}\text{C} + 32$

$$\Rightarrow F = \frac{9}{5} \times 45 + 32$$

$$\Rightarrow F = 81 + 32 = 113$$

$$\Rightarrow 45^{\circ}\text{C} = 113 \text{ F}$$

Q. Calculate 100 F in degree Celcius.

Ans

$$\Rightarrow \frac{C}{5} = \frac{F - 32}{9}$$

$$\Rightarrow \frac{C}{5} = \frac{100 - 32}{9}$$

$$\Rightarrow C = \frac{68 \times 5}{9}$$

$$\Rightarrow C = 37.78^{\circ}$$

• Some important terms related to physics

1. The smallest unit of length $\rightarrow 1 \text{ Fermi} = 10^{-15} \text{ m}$.
2. The greatest unit of length $\rightarrow 1 \text{ Parsec or, 1 parallaxic second} = 3.15 \times 10^{16} \text{ m}$.
3. 1 X ray Unit (U) $\rightarrow 10^{-13} \text{ m}$.
4. 1 Angstrom ($\text{\AA}$) $\rightarrow 10^{-10} \text{ m}$.
5. The distance between earth and sun is known as Astronomical Unit [AU] $\rightarrow 1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$.
6. 1 light year $\rightarrow 9.5 \times 10^{15} \text{ m}$.

$\rightarrow$ This is the unit of length that travelled by light in a year.

7. The smallest unit of mass $\rightarrow \text{a.m.u (atomic mass unit)}$.

$$1 \text{ a.m.u} = 1.67 \times 10^{-27} \text{ Kg} \\ = 1.67 \times 10^{-24} \text{ g}$$

8. The largest unit of mass $\rightarrow \text{Chandrashekhar Limit}$.

$$1 \text{ C.L} = 1.4 \times \text{Mass of Sun} \\ = 1.4 \times 2 \times 10^{30} \text{ Kg} \\ = 2.8 \times 10^{30} \text{ Kg}$$

9. The smallest unit of time $\rightarrow \text{Shake}$.

$$1 \text{ Shake} = 10^{-8} \text{ second}$$

10. The largest Unit of time $\rightarrow \text{Century}$.

$$1 \text{ Century} = 100 \text{ years}$$

11. The radius of earth $\rightarrow 6400 \text{ Km (Approx)}$
 $\rightarrow 6.4 \times 10^6 \text{ m}$

&

$$\text{Mass of the earth} \rightarrow 6 \times 10^{24} \text{ Kg} \\ \rightarrow 6 \times 10^{27} \text{ g}$$

12. The radius of the moon $\simeq 1738 \text{ km}$.
 $\simeq 1.738 \times 10^6 \text{ m}$

&

$$\text{The mass of the moon} = 7.34 \times 10^{22} \text{ Kg} \\ = 7.34 \times 10^{25} \text{ g}$$

13. The radius of the sun = 696000 Km.
 $= 6.96 \times 10^5 \text{ km}$.

&

$$\text{Mass of the sun} = 2 \times 10^{30} \text{ Kg} \\ = 2 \times 10^{33} \text{ g}$$

- **Atmospheric pressure**

- The pressure created by air around us is called atmospheric pressure.
- It is also called “air pressure”.

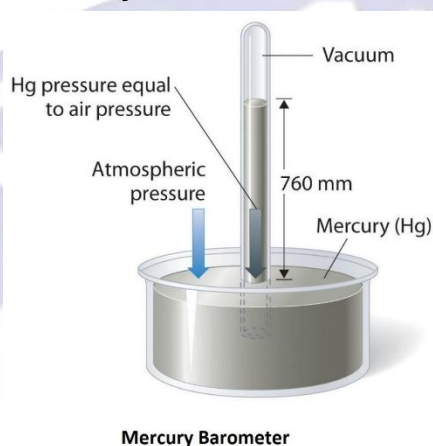
OR

The force entered on a surface by the air above it as gravity pulls it to earth.

- The atmospheric pressure is measured by “barometer”.
- The standard atmospheric pressure on the earth at sea level is 76 cm of mercury (760 mm of mercury).

Note:

1. 76 cm height of mercury = 29.92 inch



- If the reading of barometer suddenly down which indicates wind, storm and heavy rain.
- If the reading of barometer downs falls slowly then it's indicating rain.
- If the reading of barometer up slowly then it's indicating fresh weather.

- **Room temperature**

$25^{\circ}\text{C} = 298\text{ K}$ $27^{\circ}\text{C} = 300\text{ K}$
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- **S.T.P (Standard Temperature and Pressure)**

1 atm at 0°C (273 K)

Atmospheric pressure

- Any gas occupies 22.4 litre.

- **N.T.P (Normal Temperature and pressure)**

Pressure = 1 atm

Temperature = 20°C (293 K)

- Any gas occupies 22.4 litre.
- **Energy**
- The rate of work done is called “energy”.

$$\text{Energy} = \frac{\text{Work done}}{\text{Time}}$$

- The S.I unit of energy is “Joule” which is denoted by “J”.
- Another unit of energy is “Calorie”.

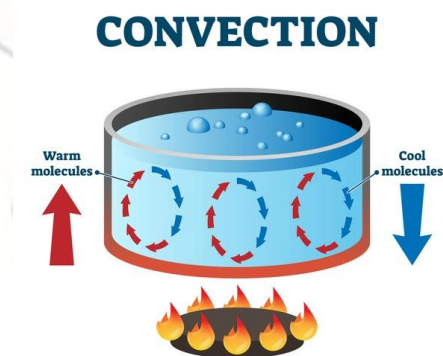
Note:

- ⇒ 1 Calorie = 4.182 Joule
- ⇒ 1 Calorie = 4.2 Joule (Approx)

- **Heat**
- Heat is a form of energy which makes the substance hot.
- Heat is a form of energy that is transferred from one body to another body.
- Heat can be transferred from one body to another body by three ways:-
 1. Convection.
 2. Conduction.
 3. Radiation.

1. Convection

- The process of heat transfer from one part of a fluid to another part by the actual movement of the particles of the fluid is called convection.
- Convection is applicable in liquid or gas.
- Other Definition: The movement of fluid molecules from higher body temperature region to lower temperature region this process is convection.

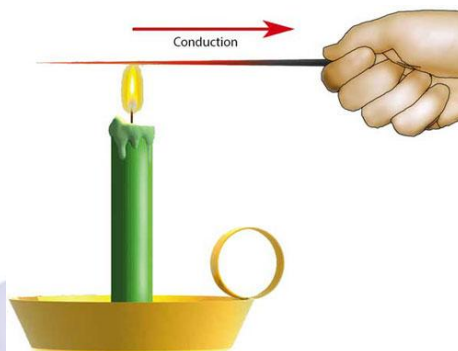


For example:-

- i) Boiling of water.
- ii) The warm air rises.
- iii) The lake, river water evaporates in the air.
- iv) Blood Circulation.

2. Conduction or Thermal Conduction or heat Conduction

- The process in which heat flows from objects with higher temperatures to lower temperature.
- It is applicable in solid substances.

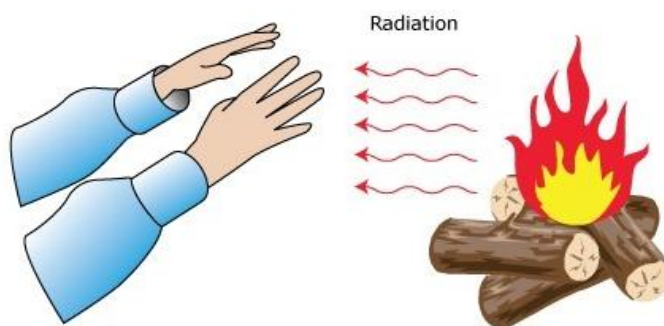


For example:-

- i) Ironing of clothes (heat is conducted from the iron to the clothes).
- ii) Heat Conduction through the sand at the beaches sand is a good conductor of heat.
- iii) Heat is transfer from hand to ice cube due to which ice melts.

3. Radiation

- Thermal radiation is generated by the emission of electromagnetic waves, these waves carry away the energy from the emitting body.
- Radiation takes place through a vacuum or transparent medium.
- Radiation is the result of the random motion of molecules in the matter.



Note:

- Heat is transferred by radiation is invisible to the eye and not affected by air flow.

For example:-

- i) Microwave radiation emitted in the oven is an example of radiation.
- ii) The release of alpha particles during the decaying of Uranium – 238 into thorium – 234 is an example of radiation.
- iii) The sun heats the ground by radiation (ultraviolet rays).